Voice Assistant for Fast Food restaurants using DistilBERT

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***Abstract*— In a time of accelerating technology development, the fast-food business is constantly looking for new ways to improve client experiences. To improve and streamline the client order-taking procedure, this research study offers a revolutionary voice-activated fast food ordering helper. We show a strong and effective voice assistant system that is capable of accurately interpreting and responding to consumer queries by using the capabilities of DistilBERT which is a BERT NLP model. This pre-trained model serves as the basis for the subsequent fine-tuning procedure, which improves performance by utilising domain-specific data from fast-food restaurant menus. User's responses and menus are stored in MongoDB which can be forwarded to the cooking section. The user's responses can be used for constantly training the voice assistant to better understand user intent. The Google-TTS and STT library converts Text to speech and speech to text respectively. After fine-tuning to DistilBERT the model gave an accuracy of 95%.**

Keywords*— *Natural Language Processing, Voice Assistant, MongoDB, Python, BERT

I. Introduction

Many restaurants are using traditional methods of waiters taking orders from customers table-to-table on pen paper and forwarding it to the kitchen. It increases crowding in tiny restaurants where people have to stand in long queues to give their orders which is inconvenient, especially for the elderly and disabled. Also, for customers with visual impairment ordering by reading the menu is a huge task where they have to depend on someone. Hiring waiters increases the capital for the restaurants since man-power is costlier.

Thus, integrating a voice assistant provides a multifaceted solution to the above and many other problems restaurants face. Now, customers no longer need to wait in line to interact with a waiter; instead, they can simply voice their orders, reducing congestion and enhancing the overall dining experience, particularly for those with mobility issues such as the elderly and disabled. Moreover, for patrons with visual impairments, navigating menus can be daunting, often requiring assistance. A voice assistant alleviates this burden by providing a spoken interface for menu exploration and order placement, promoting independence and inclusivity. Beyond regular business hours, convenience is guaranteed by its round-the-clock accessibility. Restaurants use data insights to continuously improve their products, which raises overall consumer satisfaction and loyalty in a cutthroat industry. Additionally, by replacing or augmenting human waitstaff with a voice assistant, restaurants can mitigate the rising costs associated with labor, contributing to operational efficiency and potentially reducing overhead expenses.

It also accelerates order placement and improves accuracy and customization. The confluence of voice recognition and data storage technologies in platforms like MongoDB has increased the possibilities of voice assistants in the fast-food industry even more. The data stored can be used to train VA to give more accurate results. S. Uke et al., [13] integrated a Voice Assistant into a restaurant website and used the Technology Assessment Model [TAM] to understand users' increasing need to visit a restaurant after interaction with a voice assistant. Findings indicate that the virtual agent's hedonic and utilitarian qualities greatly boost prospective customers' desire to check out the eatery. The hedonic might be important for the restaurants and hotel domains.

We used DistilBERT as our NLP to understand user intents. DistilBERT has emerged as a superior model for Natural Language Processing (NLP) due to its streamlined architecture and efficient training process. By distilling the knowledge from the larger BERT model into a smaller, more compact version, DistilBERT retains much of the original model's performance while significantly reducing computational resources and inference time. This makes it highly suitable for various NLP tasks, especially in scenarios where computational constraints are a concern, without compromising on accuracy or effectiveness. Its pre-trained weights are utilized to understand semantics related to fast-food restaurants. By utilizing pre-trained weights, our system benefits from the vast knowledge distilled from large-scale text corpora.

Our objective is to enhance customer experience and convenience as well as accuracy.

1) The customer will be able to interact with the voice assistant and give, update or delete an order.

2) The voice assistant will be able to give personalized recommendations if the customer requires them.

3) All the orders will be stored in backend in MongoDB which restaurants can utilize for improvements and understand current demand.

II. LITERATUTE SURVEY

R. Sangpal et al. [1] talks about Jarvis which is an AIML-based virtual voice assistant. The programming language used in this model is Python (3.7.2 and 3.6.8). Jarvis uses the speech-to-text (STT) and text-to-speech (TTS) technology from Google's GTTS. The authors in the paper have recognized 3 generations of VAs and the paper focuses on the 3rd generation where systems are built using XML. The paper [12] shows tells about the storage of user data in the cloud by intelligent virtual assistants (IVAs) like Amazon Alexa and Google Assistant. It underscores the potential privacy and security threats associated with this practice, outlining the types of data stored by Alexa, including traffic card, sport card, shopping item, task, eon card, notification card, and puffin card, emphasizing the implications for user privacy and provides suggestions for users and vendors to address privacy concerns. [2] This paper focuses on the privacy of the data recorded by the Virtual Assistant. The use of any VA depends on 1) the Quality of speech recognition and 2) VA interaction. The conjoint model of virtual assistants is described in depth in this paper. The paper says that the answers given by the VAs are always personalized and contextual depending on the search history of the web pages. [3] Voice User Interface (VUI) enables an intelligent personal assistant to carry out mental tasks, such as turning on and off smartphone applications. Three things make up a voice assistant: an interface, dialog management, and an input/output medium for communication.

[4] VAs have significantly improved the effectiveness and user experience of virtual assistants through advancements in natural language processing and machine learning. [5] This paper proposes a new framework for film-based hand gesture recognition that overcomes this limitation. The framework utilizes real-time video acquisition, video normalization, and a novel Hand Frames Feature Extraction (HFFE) stage. [6] The study focuses on creating prototypes for voice-controlled digital banking and online payments with two-factor authentication. The research explores the potential of utilizing cloud technology and devices equipped with voice assistant software, particularly Amazon's Alexa, for secure authentication. [7] To create lightweight, speech- and vision-enabled virtual assistants for smart home and automation applications, this article provides a software architecture. The architecture consists of modules for face detection, face recognition for user identification, and speech synthesis and recognition. An application prototype named. [8] This research introduces three new scales specifically designed to measure the user experience (UX) of voice assistant systems. These scales are intended to be used alongside a modular questionnaire concept to effectively evaluate voice user interfaces (VUIs). [9] Smart virtual assistants have the potential to be applied as AI-driven laboratory assistants. They could assist in tasks such as reading standard operating procedures, reciting chemical parameters, and reading out laboratory devices and sensors. The authors conclude that VUIs have great potential in laboratory settings for instrument control and retrieving and saving experimental data and protocols.[10] This paper presents a solid framework that leverages cloud storage and blockchain to secure medical data in Healthcare 4.0. Elliptic Curve Digital Signature Algorithm is used for authentication, and Elliptic Curve Cryptography is used for encryption. The framework addresses issues with medical data security and privacy while improving security and outperforming current solutions.

[11] The adoption of voice assistants (VA) addresses the shortcomings of conventional models in comprehending AI's emotional impact. It introduces "artificial autonomy" as a crucial quality, which includes autonomy in detecting, thinking, and acting. [13] Virtual Personal Assistants (VPAs) have evolved in the age of Artificial Intelligence and Machine Learning, here we will explore the creation of a Python-based voice assistant for Windows that aims to improve accessibility and usability, with a focus on hands-free operation and email functionality.[14] This paper proposes a digital restaurant system to minimize infection risks for customers and personnel. Utilizing contactless technologies, digital menus, online ordering, and automated payment systems, the solution enhances safety and efficiency, reducing physical interactions and potential contamination in dining environments.
[15] This paper introduces QARMA algorithms for Industry 4.0, emphasizing their computational performance, predictive accuracy, and explainability over traditional ML and Deep Learning models. Practical field deployment and validation on two production lines highlight these advantages, supported by a state-of-the-art Industrial Internet of Things platform. [16] This paper tackles communication challenges for the hearing impaired by developing a gesture recognition system using image classification models. Leverages computer vision and image processing to capture and interpret. [17] This research proposes an enhanced hand gesture recognition (HGR) system using an optimized Artificial Neural Network (ANN). Traditional ANNs struggle with dynamic gestures. To address this, the authors introduce a new approach that combines an enhanced flashfly optimization algorithm with a standard ANN.

[18] In this paper for those who have hearing loss, this creates computer vision models for machine learning that can precisely record and understand hand movements. It looks at different image categorization techniques and finds the best one. A real-time video model recognizes commonly used signs, improving communication and providing a wide range of uses. [19] Glass-Box is a voice-activated gadget that uses declarations of class-contrasting

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counterfactuals to explain automatic decisions. Through interactive discussion, it draws attention to biases and flaws in decision-making models, rendering the process visible and easily comprehensible for non-expert users. Taking on the role of a loan applicant is one way to demonstrated in the paper on how automated decision-making works. [20] This paper proposes a Text-to-Speech synthesizer that uses Digital Signal Processing (DSP) and Natural Language Processing (NLP) to translate text into spoken speech. After text is processed by the application, synthesized voice is produced, played aloud to the user, and stored as an MP3 file.

III. METHODOLOGY

We used a thorough fine-tuning procedure to adapt our voice-activated fast-food ordering assistance to the particular needs of the domain. This helps to improve performance and a better understanding of the language and context to fast-food restaurants. The model is also able to understand domain-specific words and slang and deliver accurate responses thus, giving a better user experience. The model was trained on a dataset containing 10,000 entries of restaurant menus, ingredient descriptions, and a wide range of customer interactions, including order placements and customization requests, from a restaurant in the USA.

A. *Importing required libraries*

Several libraries are needed to import for required functionalities.

1) “Speech_recognition” for converting customers' speech to text and “GTTS” for converting text to speech.

2) The “Os” operating system library helps to perform operating system-related functionalities, we use it here to store audio responses from voice assistants in local storage if required.

3) “Numpy”, and “transformer” to tokenize the input text.

4) “Time” to tell the current time.

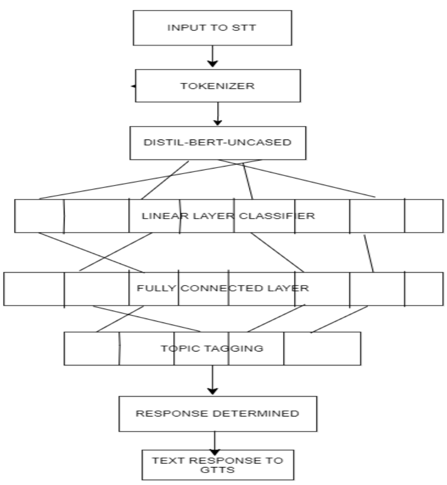

Fig 1. Specific Architecture of Virtual Voice Assistant

Let's delve into the architecture of Voice Assistant

- Input to STT**:** After hearing the customer’s response in speech it is converted to text Speech-to-text library.
- Tokenizer**:** DistilBERT requires input in a tokenized format which is usually done using IDs, and masks. This helps to capture the essence of the input text.
- DistilBERT uncased**:** The tokenized text is passed to DistilBERT which has already learned a great deal from studying loads of text data. Using what it's learned, it processes the text you've given it, trying to understand its meaning.
- Linear Layer Pre-classifier: Once DistilBERT understands the input text, it focuses on picking out the most important parts. It harnesses the enriched hidden state output, channeling it through a series of computations within a fully connected layer.
- Fully Connected Layer (Making a Decision): This layer is used to predict the intent based on different classes (words). This helps to predict the probability of an intent according to the input text.
- Topic Tagging: Based on the different probabilities obtained from previous layer, the final intent is determined in this layer.
- Response Determined: This intent is passed to our Python backend. Based on the intent response is determined.
- Text Response to GTTS: The text response determined is sent to Google text-to-speech library which converts it to audio and sent to frontend using Gradio API.

During its operation, DistilBERT processes the text through these layers. It takes the input, understands it, picks out the important bits, and then makes a decision based on that information. This enables it to provide valuable insights and predictions, such as determining the sentiment of a product review.

B. *Flow of Voice Assistance*

1) Firstly, we have to choose the right NLP model and algorithm to train the dataset for intent classification. Distil-BERT is a small yet powerful NLP model to train the dataset.

2) After training the model it is stored in a portable pickle file format, enabling seamless loading and utilization across diverse computing environments.

3) After the training process, we make a voice assistant which sends text to the model file. The user will speak into the voice assistant and the speech will be converted to text using the Speech to Text(STT) library.

4) After sending the model predicts the user’s intent.

5) There are 5 intents defined in this project

6) *food_order.name.item*

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food_order.other_description.item

food_order.type.food

food_order.time.pickup

food_order.num.people

7) The Model sends the intent back to voice assistant according to which the chatbot determines the response.

8) The response in text form is converted to speech using the Text-to-speech (GTTS) library.

9) The output is given to the user [Figure 5: Final output showcased on the display screen along with the audio.]

10) Once detected, the intent is linked to the user's input. If the purpose is to place an order or do an actionable task, the ChatBot records the interaction, together with the identified intent and a timestamp, in a MongoDB database. This logging procedure ensures that all user interactions and intents are systematically captured for future study or reference.

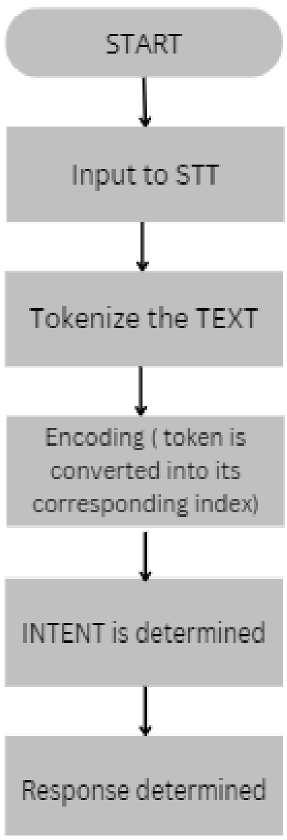

Fig 2. Systematic Representation Flowchart of VA dataflow.

IV. EQUATIONS

Accuracy: This measures the model's ability to correctly predict the outcome or label for a given input text. In eq(1), accuracy is measured for our fine-tuned model on DistilBERT. During the evaluation process, each predicted intent is compared against the corresponding correct intent label. If the predicted intent matches the ground truth label, it is considered a correct prediction. The accuracy is then calculated as the percentage of correct predictions out of the total number of predictions made by the model.

$$A_C = \frac{N_C}{TN} \cdot 100 \qquad \text{--- (1)}$$

A_C : Accuracy
Nc : Number of correct predictions
TN: Total number of predictions

Training Loss: The training loss represents the average loss incurred over all batches during the training process. It is calculated by summing up the losses for each batch and dividing by the total number of batches as shown in eq(2).

$$T_\gamma = \frac{\sum_{i=1}^{N} L_i}{N} \qquad \text{---(2)}$$

T_γ : Training Loss
N: Total Number of batches
L_i: Loss for i^{th} batch.

V. RESULTS

The model gave an accuracy of 95%, The metric is derived from evaluating the model's predictions against a labeled dataset during the training phase the accuracy is calculated by comparing the model's predicted intents with the ground truth labels for a set of test examples. These test examples consist of the same intents as were given in the training set. A training loss of 0.676 over the testing set is measured which is relatively low indicating that our model can accurately predict any future unforeseen statement with low error. We were able to accurately read and answer consumer queries about food orders by utilizing the capabilities of DistilBERT, a tiny yet powerful NLP model. Above is the interface our user sees while interacting with the voice assistant. "You said" is the customer input in text while, "AI said" is the response in text and audio as well.

The data user responses are stored in the backend in MongoDB which can be used for further training of the model thus helping it to give more human-like responses by training on the new dataset reloading it into a pickle file and including it in the voice assistant, and restaurants can also analyze the customer needs and food demands.

The study's findings show that a voice-activated fast food ordering assistant can be successfully implemented

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by utilizing cutting-edge natural language processing (NLP) approaches.

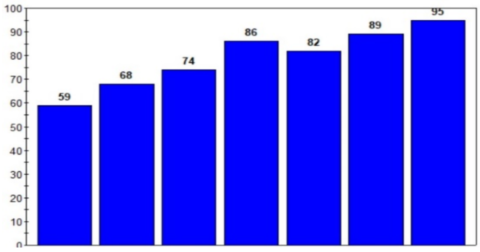

Fig 3. Accuracy over 7 epochs

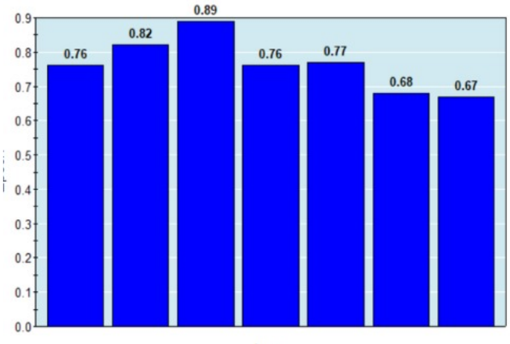

Fig 4. Training loss over 7 epoch

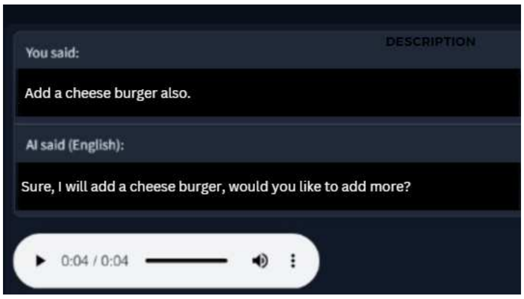

Fig 5. The final output is showcased on the display screen along with the audio.

Above is the interface our user sees while interacting with the voice assistant. “You said” is the customer input in text while, “AI said” is the response in text and audio as well.

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The study's findings show that a voice-activated fast food ordering assistant can be successfully implemented by utilizing cutting-edge natural language processing (NLP) approaches.

It achieved a simplified and efficient interaction process between the user and the voice assistant by employing our methodology, which included tokenization of user inputs, intent detection using the trained model, and answer production.

VI. Future Scope

The field of AI, NLP has a vast future scope in the food industry. Let us check the future scope of virtual assistants:

• The virtual assistant can be the restaurant’s backend to store and send data directly to the cooking section.

• It can be integrated with cloud platforms for a single virtual assistant to handle various other restaurants.

• The data collected from Virtual Assistant can be used by restaurants to change menus according to customer needs.

The virtual assistant can be used in various places to automate tasks, answer repetitive and standard question answers, and other tedious tasks.

• It can be used in apps and websites to create personalized smart applications.

• Can be used in hospitals to give patient data to users.

• In weather forecasting give daily weather updates according to place, time, and date.

VII. Conclusion

Through the utilization of machine learning (ML) and natural language processing (NLP) tools, we developed a sophisticated Virtual Assistant (VA) that streamlines the food ordering process while significantly reducing the need for manpower. Our choice of employing DistilBERT enables data training even in resource-constrained environments and optimizes the performance of our model. Our VA integrates user inputs, processes them through NLP algorithms to discern intent, and generates tailored responses accordingly. Leveraging MongoDB, we efficiently store customer orders and interactions, facilitating not only real-time order management but also serving as valuable data for ongoing model refinement and customer demand analysis.

The versatility of our VA extends beyond mere digital interfaces, as it can be integrated with IoT devices such as speakers, allowing for its deployment in diverse environments for enhanced accessibility and convenience. In essence, our smart VA represents a union of technologies, empowering businesses to streamline operations, enhance customer experiences, and adapt dynamically to evolving market demands.

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